SilentBell

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Introduction

SilentBell is a smart, silent doorbell system designed for individuals with hearing impairments.
When a visitor presses the doorbell button, the system sends a signal to an Arduino microcontroller, which then activates multiple non-auditory alerts. These include a Wi-Fi–controlled smart LED bulb, wired LED light strips inside the home, and instant push notifications to the user's smartphone and smartwatch.

Methods

1. Hardware Workflow

Doorbell Press (12V Input)

- Visitor presses the push-button powered by a 12V adapter.
- The output passes through a 10 kΩ + 3.3 kΩ voltage divider, safely reducing it to 3.3V for the ESP32 input pin.

Signal Detection (ESP32)

- ESP32 reads the 3.3V logic signal on GPIO D2.
- Software debouncing removes noise and prevents duplicate readings.

Activation of Alerts

Once the ESP32 detects a button press, it simultaneously triggers:

- LED Strip: Powered by its own 5V adapter and controlled directly using GPIO D4.
- Wi-Fi Smart Bulb: ESP32 sends Wi-Fi API commands to turn on/change color.
- Mobile + Smartwatch Notification: Sent instantly to the user's device.

2. Power System

- 12V Adapter → Doorbell Button
- 5V Adapter → LED Strip
- ESP32 powered via USB Type-C
- Common GND shared across all components

3. Software & App

- ESP32 programmed using Arduino IDE
- Wi-Fi commands for smart bulb via cloud/local API
- Mobile app receives push notifications
- Smartwatch mirrors mobile notifications

Results

•The system successfully combined a smart light bulb, an LED strip, and a mobile app into one easy-to-use notification device for the home.

•The doorbell button works by instantly detecting when someone presses it and triggering the alert system.

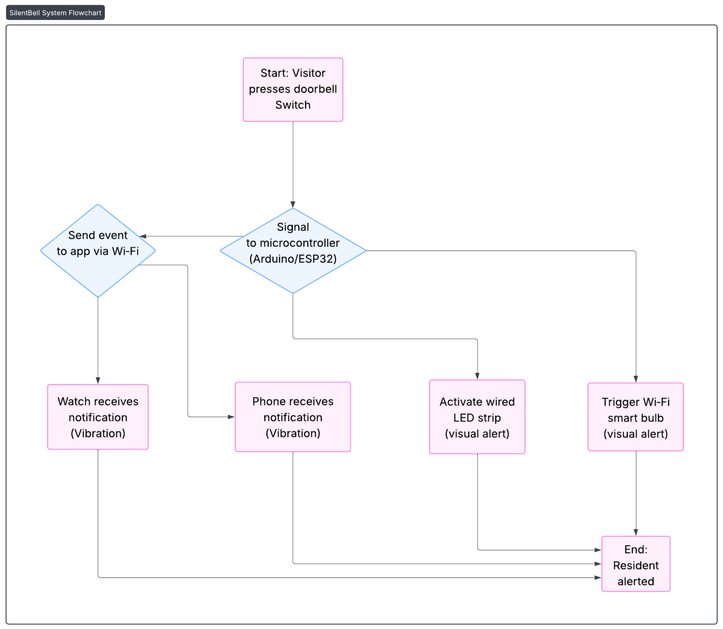
•When the doorbell is pressed, the smart light bulb automatically connects to the Wi-Fi and flashes a bright, colorful pattern to visually alert anyone in the room.

•Users receive instant notifications on both their smartphones and smartwatches, so they know someone is at the door even if they aren't looking at the lights.

•Every time the doorbell rings, the event is securely recorded in the cloud, giving the user a complete history of visitors for safety and peace of mind.

•The system is designed to be reliable and will automatically reconnect to the internet after a power outage or network issue, so it's always ready to work.

Flowchart:



Results Cont.

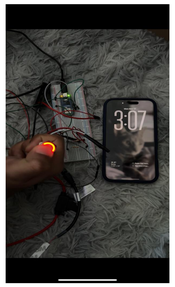


Figure 2.1 The LED when the button is pressed

Figure 2.2 The real time push notification on screen

The entire workflow, from the physical button press to the wireless transmission and final reception by the user's phone is shown in the result's demo

Discussion

SilentBell effectively replaces sound-based doorbells with non-auditory alerts for accessibility. The voltage-divider approach allowed safe interfacing between a 12V button and the ESP32 without additional ICs. Using both a Wi-Fi bulb and LED strip gives redundancy, if a phone alert is missed, visual cues still notify the user. The system met usability goals: fast response, simple installation, low cost, and dependable operation. Future improvements include battery backup, camera integration, and multi-color alert patterns for different visitors or events.

References

- ANSI. *Accessible and Usable Buildings and Facilities (ANSI A117.1)*. American National Standards Institute, 2023.
- California Legislature. *California Consumer Privacy Act (CCPA)*, Cal. Civ. Code § 1798.100, 2024.
- European Parliament. *General Data Protection Regulation (GDPR)*, Regulation (EU) 2016/679, 2016.

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